

CO3_AT02_Interactive Tree simulator

<https://sensational-sundae-a90704.netlify.app/>

The screenshot displays the 'CO3_AT02_Interactive Tree simulator' web application in a browser. The interface is divided into three main sections: 'BUILD THE TRIE', 'LIVE STRUCTURE', and 'STATS'.

BUILD THE TRIE (Panel 01): This section allows users to interact with the trie. It includes an 'Add a word' input with an 'Add' button, a 'Load example words' button, a 'Look up' section with 'Whole word' and 'Prefix only' tabs and a 'Check' button, and a 'Remove a word' section with a 'Remove' button. A 'Clear everything' button is at the bottom.

LIVE STRUCTURE (Panel 02): This panel shows the current state of the trie, labeled '7 words stored'. The structure is a tree starting from a root node (a diamond). The root has two children: 'C' and 'D'. 'C' has a child 'A', which has children 'R' and 'T'. 'R' has children 'D' and 'E'. 'D' has a child 'O', which has children 'G' and 'T'. The nodes are represented as boxes with letters, and the edges are lines connecting them.

STATS (Panel 03): This panel provides statistics about the trie. It shows '7 words stored', '11 nodes (incl. root)', '4 max depth', and '22 letters typed in'. A message states: 'You typed 22 letters in total, but the trie only needed 10 letter-nodes to store them — 55% reused via shared prefixes.' Below this is a 'TRY A PREFIX' section with a 'start typing...' input and a 'completions appear here' output. At the bottom is a 'WHAT'S HAPPENING' section with a log of actions: '- Adding "DOT" -', 'D - already exists here, reusing the branch', and 'O - already exists here, reusing the branch'.